

Combined AI Findings Report

Milestone 2: AI-Assisted Network Traffic Threat Analysis

Section 1: Investigation Overview

This report consolidates the AI-assisted analysis performed across three PCAP-derived CSV datasets extracted from the same underlying capture, 2021-08-16-formbook.pcap: traffic_segment_1.csv, traffic_segment_2.csv, and traffic_segment_3.csv. Each dataset was reviewed independently before being brought together here so that patterns spanning multiple segments could be identified and cross-referenced.

The objective across all three segments was consistent: identify potentially suspicious network behaviour, communication patterns, external relationships, protocol usage, and any indicators that would justify deeper security investigation. ChatGPT was used as an AI-assisted analysis tool to generate an initial set of security observations directly from the network metadata contained in each CSV export.

Every AI-generated finding was reviewed from a cybersecurity analyst perspective before being included in this document. None of the observations here are treated as confirmed threats on their own; each is presented as an indicator that still requires validation through packet payload inspection, endpoint investigation, or threat intelligence lookups before it can inform an incident response decision.

The underlying CSV files were produced by converting the raw PCAP capture into structured records containing source and destination IP addresses, ports, protocols, packet counts, and frame lengths. Working from CSV rather than the raw capture allowed the AI tool to reason over the traffic at a metadata level; it did not have access to packet payloads, so none of the findings below should be read as payload-level confirmation of malicious activity.

Section 2: Dataset Summary

The three datasets cover a combined total of 3,588 packets captured during the same session.

Segment	Dataset	Packets Analysed
Segment 1	traffic_segment_1.csv	1,200
Segment 2	traffic_segment_2.csv	1,200
Segment 3	traffic_segment_3.csv	1,188

Each segment was extracted and analysed as a discrete unit, which is why the packet totals differ slightly between segments despite originating from the same source capture. Segment 3 was reviewed after correcting formatting issues found in the initial CSV export for that portion of the capture, so its record count reflects the corrected dataset rather than the original one.

Section 3: Common Internal Host Identified

Primary Internal Host Across Investigation

Across all three traffic segments, the internal IP address 10.10.0.33 was consistently identified as the primary communicating internal endpoint. In every segment, this host appeared as both a major source and a major destination of network traffic, which indicates it was the main system involved in outbound communication to the internet during the capture window.

Observation	Evidence
Internal host	10.10.0.33
Classification	Private IP (RFC1918)
Role	Main communicating endpoint
Investigation priority	High

Because 10.10.0.33 falls within the RFC1918 private address range (10.0.0.0/8), it is confirmed to be an internal host rather than a public-facing system. The consistency of this host's presence across three separate segments, captured at different points in the same session, is the single strongest signal in this investigation: it points to one system generating the bulk of the traffic that requires review, rather than a spread of activity across the network. For that reason, 10.10.0.33 is assigned the highest investigation priority of any asset identified in this analysis.

Section 4: Major External Communication Patterns

Repeated External Communication Relationships

The analysis identified repeated communication between the internal host 10.10.0.33 and several public internet addresses. The most significant relationships observed in each segment were:

Segment 1: 10.10.0.33 <-> 18.184.26.60
Segment 2: 10.10.0.33 <-> 18.184.26.60
Segment 3: 10.10.0.33 <-> 204.79.197.200

External IP	Segments Observed	Observation
18.184.26.60	1, 2	Highest communication volume
204.79.197.200	2, 3	Repeated HTTPS communication
88.221.0.167	1	Large external response traffic
8.8.8.8	1, 2, 3	DNS communication

18.184.26.60 stands out as the most significant single relationship in the investigation. It accounted for the highest communication volume in both Segment 1 and Segment 2, which suggests either a

sustained connection or a repeated pattern of connections to the same external service across two separate portions of the capture. 204.79.197.200 became the dominant external relationship in Segment 3 and also appears in Segment 2, carrying what was identified as repeated HTTPS communication. 88.221.0.167 was notable in Segment 1 for large external response traffic, and 8.8.8.8, Google's public DNS resolver, appears consistently across all three segments, which is expected for a host performing routine domain resolution.

None of these external addresses can be classified as malicious from metadata alone. Each relationship is flagged here because of its volume or its recurrence across segments, not because of any confirmed indicator of compromise. Establishing ownership, reputation, and purpose for each of these addresses is one of the priority items in the recommendations set out in Section 10.

Section 5: Protocol Analysis

Observed Network Protocols

The investigation identified several common network protocols across the three segments: TCP, HTTP, HTTPS/TLS, DNS, NBNS, SSDP, and ARP. The majority of communication used TCP-based services, with HTTP and HTTPS accounting for the largest share of application-layer traffic.

Protocol	Security Interpretation
HTTP Port 80	Unencrypted web communication requiring review
HTTPS Port 443	Encrypted web/application traffic
DNS Port 53	Normal domain resolution
NBNS	Internal Windows discovery traffic
SSDP	UPnP discovery traffic

HTTP traffic on port 80 is unencrypted, so any credentials, session tokens, or application data carried in that traffic would be exposed to anyone able to observe the network path; this traffic is flagged for review in every segment where it appears. HTTPS traffic on port 443 is encrypted at the transport layer, which limits what can be inferred from metadata alone but is also the expected behaviour for most modern web and application traffic. DNS traffic on port 53 is consistent with normal domain resolution.

NBNS and SSDP are both local discovery protocols: NBNS supports legacy Windows name resolution, and SSDP supports UPnP device discovery. Neither is inherently a security concern, though both can indicate services running on a host that a hardened environment would typically disable. ARP traffic reflects normal local network address resolution and was noted rather than flagged.

Section 6: Beaconing and Automated Behaviour Assessment

Automated Communication and Beaconing Analysis

Repeated communication patterns were observed throughout the investigation, particularly in the relationship between 10.10.0.33 and its top external peers in each segment. Given how often this kind of pattern is associated with malware command-and-control channels, each segment was assessed specifically for indicators of beaoning behaviour.

The traffic showed several characteristics consistent with automated activity: high packet repetition between a small number of host pairs, communication concentrated into short windows rather than spread evenly across the capture, and exchanges that looked automated rather than manually driven. The traffic did not, however, show the characteristics most closely associated with classic malware beaoning: there was no evidence of regular, periodic callbacks at fixed intervals, and no traffic pattern consistent with known command-and-control protocols.

Based on this combination of factors, beaoning is assessed at low confidence across all three segments. This is not a clean bill of health: automated-looking traffic without confirmed periodicity can still represent malicious activity that simply does not fit the classic beaoning model, particularly with modern malware families that deliberately randomize their callback timing to avoid detection. Confirming or ruling out beaoning definitively would require PCAP payload inspection to examine the actual content of the exchanges, DNS analysis to check for domain generation algorithm patterns or unusual query behaviour, and endpoint investigation to identify the process responsible for the traffic.

Section 7: Data Transfer Analysis

Large Data Exchange Observations

Several larger data exchanges were observed between the internal host and external systems across the three segments. Examples include sizeable external web responses, ongoing application communication, and traffic patterns consistent with possible file downloads.

Packet metadata alone does not confirm unauthorized transfer or data exfiltration in any of these cases. Large transfers are common in legitimate contexts: web page assets, application updates, and cloud service synchronization can all produce transfer volumes similar to what was observed here. What makes these transfers worth flagging is less their size and more the context around them, namely that they occur as part of the same concentrated relationships identified in Section 4. Confirming the nature of these transfers requires inspecting the actual payload content, which was outside the scope of the metadata-level analysis performed here.

Section 8: Security Concerns Identified

Finding	Confidence
Heavy communication with external services	Medium
Possible automated application behaviour	Low-Medium

HTTP unencrypted traffic	Medium
Malware communication	Low
Command-and-control activity	Not confirmed
Data exfiltration	Not confirmed

These confidence levels reflect how strongly the available metadata supports each finding, not a determination of actual risk. A medium confidence rating on heavy communication with external services means the pattern is clearly present in the data, not that it has been confirmed as malicious. Command-and-control activity and data exfiltration are both listed as not confirmed rather than assigned a numeric confidence level, since the investigation found no direct evidence either supporting or ruling out either possibility at the metadata level.

Section 9: Confirmed Observations

The following observations were confirmed directly from the datasets and are not dependent on interpretation:

- Internal host 10.10.0.33 identified as the primary communicating endpoint
- Multiple external internet communications observed across all three segments
- TCP, HTTP, HTTPS/TLS, and DNS traffic identified and classified
- Repeated communication relationships detected between 10.10.0.33 and a small set of external addresses
- Large data exchange patterns observed
- No confirmed malware activity identified in any of the three segments

Section 10: SOC Analyst Recommendations

The following steps are recommended before this activity can be conclusively classified as benign or malicious:

1. Perform threat intelligence checks on external IP addresses. Run 18.184.26.60, 204.79.197.200, 88.221.0.167, and the other public addresses identified across the three segments through reputation platforms such as VirusTotal and AbuseIPDB, and confirm ownership through WHOIS and ASN lookups.
2. Extract DNS queries and review requested domains. Identify what domains 10.10.0.33 resolved during the capture window, and check for indicators such as unusually long or randomized domain names that could point to DNS tunnelling or domain generation algorithm activity.
3. Inspect TCP streams and packet payloads. Reconstruct the sessions between 10.10.0.33 and its top external peers to determine what data was actually exchanged, rather than relying on packet counts and byte totals alone.
4. Identify the process responsible for communication from 10.10.0.33. Endpoint tooling should be used to map the observed network activity back to a specific running process or application on the host.

5. Compare traffic against normal host behaviour. Establish a baseline for 10.10.0.33 using historical traffic data, if available, to determine whether the volume and pattern of communication observed here falls inside or outside its normal range.
6. Review endpoint security logs and EDR alerts. Check whether any antivirus, EDR, or host-based monitoring tooling on 10.10.0.33 has already flagged activity coinciding with the traffic observed in these three segments.

Section 11: Final Analyst Conclusion

The combined AI-assisted analysis identified 10.10.0.33 as the primary network participant across all three investigated segments. The host demonstrated repeated communication with multiple external systems, including high-volume relationships with 18.184.26.60 in Segments 1 and 2 and with 204.79.197.200 in Segment 3.

Several of the patterns identified here, concentrated external communication, unencrypted HTTP traffic, and automated-looking exchange behaviour, warrant further investigation. That said, the available network metadata does not provide sufficient evidence on its own to classify this activity as malicious. No confirmed malware infection, command-and-control channel, or data exfiltration was identified in any of the three segments.

Further validation using packet payload analysis, endpoint investigation, and threat intelligence sources is required before a final security determination can be made. Until that validation is complete, 10.10.0.33 should remain the top investigation priority arising from this milestone, and the recommendations in Section 10 should be treated as the next steps in that process.